A Hypertension Parallel Healthcare System Based on the ACP Approach

Lin Ge, Bo Lv, Na Li, Shan An, *Senior Member, IEEE*, and Fei-Yue Wang, *Fellow, IEEE*

***Abstract*—Hypertension, the most common chronic disease, is an enormous risk factor for cardiovascular and cerebrovascular diseases and vascular events, which has become a significant global public health problem. However, the etiology and pathogenesis of hypertension are incredibly complex. In conventional medicine, the intervention and treatment focus on the blood pressure (BP) indicator. At the same time, the Traditional Chinese Medicine (TCM) theory provides diverse perspectives, more indicators, and optional treatment. In addition, the parallel intelligence theory based on the artificial systems, computational experiments, and parallel execution (ACP) approach is proper research and practical methodology for solving management and control problems for the treatment of hypertension. In this paper, we proposed a parallel healthcare system (PHS) framework to describe patients' conditions and diagnoses, trace the disease evolution progression, and analyze and optimize available therapeutic regimens. At last, we constructed a PHS for hypertension under TCM theory which applied to the treatment of actual volunteered patients and achieved positive effects.**

***Index Terms*—Hypertension, parallel intelligence theory, ACP approach, parallel healthcare system.**

I. Introduction

HYPERTENSION is a considerable risk factor for stroke, cardiovascular disease, and vascular events and is one of the most significant contributors to the burden of death and disability worldwide [1]. Raised BP affects the structure and function of the heart, brain, kidneys, eyes, and other essential organs, ultimately leading to organ failure. The complexity of hypertension brings various challenges to the patient's individualized treatment and long-term health management. Despite development in research, the most common treatment is prescribed single or combined anti-hypertensive drugs for long-term even lifetime, which the patient is not satisfied. The TCM mainly follows the pathogenesis and syndromes and treats patients according to clinical symptoms. It provides treatment methods to stabilize symptoms, reduce risk factors, protect target organs so that the patients achieve the purpose of drug withdrawal and dosage reduction, and improve their quality of life [2]. It has shown that TCM is particularly valuable in chronic hypertensive disease management and treatment.

The human body is a complex system with multiple influence factors and different pathogenesis for hypertension. Therefore, the key to treatment is identifying the etiology, precise medication regimen, personalized lifestyle changes, and dynamic treatment plan adjustment [3]. The existing research predominantly concentrates on the standard of diagnosis, optimization of drug therapy strategy, and healthy lifestyle recommendations. However, few studies focus on personalized pathogenesis and precise treatment for individuals. The ACP approach is a practical theory and method for studying complex systems by expanding the physical system into an artificial system and implementing computational experiments with extensive data [4], [5]. The TCM is well known for treating the human body as a whole system, not as separated organs or tissue, and provides comprehensive intervention and more optional programs than pharmacological treatment. This paper aims to introduce a PHS for hypertension treatment based on the ACP approach using the TCM method to provide efficient, economical, and personalized treatment. With the combination of APC and TCM, we can determine the possible pathogenesis of the disease and the best treatment for patients through real-time symptom tracking, fastidious estimation, and treatment optimization.

The remainder of this paper is structured as follows: Section II introduces a PHS for hypertension treatment and its framework. Section III discusses the principle of TCM, which reveals the pathogenesis, etiology, and critical parameters. Section IV presents a case study of actual patients' treatment using PHS. Finally, Section V concludes the current work achievements and offers future work issues.

II. Framework of Parallel Healthcare System For Hypertension

Parallel intelligence based on the ACP approach is a valuable methodology that solves uncertainty, diversity, and complex problems [6]. With the advantages of this methodology, PHS focuses on the research of the complex human live system, determination of the pathogenesis of the disease, and assistance for the doctor in providing optimized treatment schemes. From the

Manuscript received 26 July 2022; revised 20 September 2022; accepted 6 October 2022. Date of publication 17 October 2022; date of current version 12 December 2022. This work was supported in part by the Research and Development Programs by Sungohealth Inc. *(Corresponding author: Lin Ge.)*

Lin Ge is with the Faculty of Innovation Engineering, Macau University of Science and Technology, Macau, China (e-mail: 2109853pmi30002@student.must.edu.mo).

Bo Lv and Na Li are with the Institute of Life Science, Sungohealth Inc., Beijing 100055, China (e-mail: lvbo@sungohealth.com; lina@sungohealth.com).

Shan An is with the School of Computer Science and Engineering, Beihang University, Beijing 100091, China (e-mail: anshan.tju@gmail.com).

Fei-Yue Wang is with the State Key Laboratory for Management and Control of Complex Systems, Chinese Academy of Sciences, Beijing 100080, China, and also with the Faculty of Innovation Engineering, Macau University of Science and Technology, Macau, China (e-mail: feiyue@ieee.org).

Digital Object Identifier 10.1109/JRFID.2022.3214867

2469-7281 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Padre Conceicao College of Engineering. Downloaded on August 06,2024 at 09:21:45 UTC from IEEE Xplore. Restrictions apply.

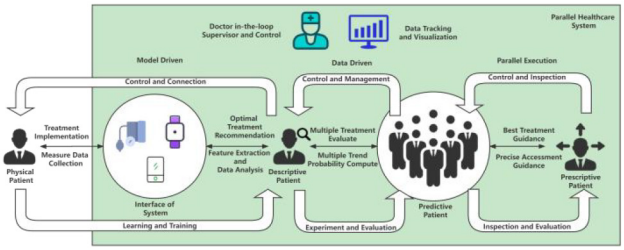

Fig. 1. Framework of Parallel Healthcare System.

modeling perspective, the PHS mainly consists of four software-defined patient models: the physical patient, descriptive patient, predictive patient, and prescriptive patient [7].

The framework of PHS is shown in Fig. 1. The physical patient is the human body living system, which contains the following parameters: (a) static parameters such as clinical evidence, medical history, and living habits, (b) dynamic parameters such as physiological and psychological activity, and continuous measurable parameters, (c) external parameters such as circumstance temperature, ambient noise, environmental conditions. The patient-system interface includes smart wearable devices, electronic blood pressure monitors (sphygmomanometer), smart cellphones, and apps to collect parameter data that bridge the physical and descriptive patient. The description patient is a digital representation in the virtual world with the same category parameters, which implies the physical patient's status in the real world. Both evolve continuously and equally, following the same medical and life sciences rules. The prediction patients evolved from the descriptive patient through computational experiments with variations and perturbations of the parameters, which indicate all the probable assessments, available treatment effects, and developing trends for the disease. Under experienced doctors' supervision, the prescriptive patient evaluates the most likely diagnosis and the best treatment by inspecting prediction patients to make the eventual decision. Finally, the physical patients receive medication prescriptions, therapeutic regimens, and doctors' instructions from the PHS.

The development of disease is a continuous dynamic process. The measurement collects static, dynamic, and external data for the description patient at a given moment. Then, the descriptive data are used to evaluate the patient's status, such as clinical state, severity level, and concurrent disease. According to the evaluation result, the description patient will propose an optional treatment recommendation. Thus, the description patient at one specific moment comprises measurement, evaluation, and treatment, which can be soft-defined and objectively represent the health or disease status. As shown in Fig. 2, one patient's description at a specific time consists of a measure-evaluate-treat discrete fundamental process. The time sequence of descriptions illustrates the continuous disease progression and clinical treatment effect as the parameters change along with the disease development. The description intelligence deserves from various patients' descriptions denotes all probable disease progression and potential clinical treatment pathways.

The prediction patients come from the computational experiments of description patient, including feasible assessment,

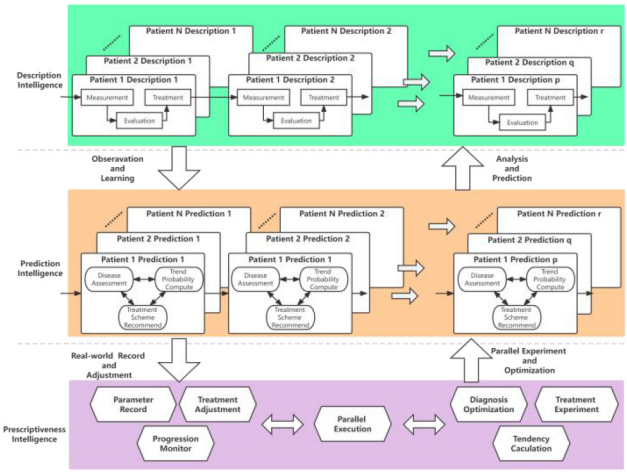

Fig. 2. The procedure of computational experiments for Parallel Healthcare System.

potential treatment schemes, and possible clinical outcome experiments, which calculate the likely diagnoses, optimized treatment plans, and probable post-treatment effects under the integration of TCM and evidence medicine. There is a one-to-many mapping relationship between description patient and prediction patients. Series sets of predictions in the chronological sequence correspond to the series descriptions. In addition, mass computational prediction experiments generate the prediction intelligence to improve diagnosis accurately, optimize treatment strategy, and guide patients' treatment.

When the parallel execution of descriptions and predictions interact, the prescriptiveness intelligence emerges [8]. On the one hand, descriptions and measurable biological indicators of the patient imply the disease status and the implementation of treatment. On the other hand, predictions and computational experiments help doctors to recommend the best therapeutic regimen and forecast the effect of treatment. The prescriptive patients request a recalculation of the diagnosis and adjusting treatment scheme timely if the deviation between descriptions and predictions is remarkable.

In brief, the PHS based on the ACP approach illustrates a virtual-real interaction paradigm between the physical and computational patient to provide optimized, personalized medicine and health management services. Meanwhile, under the current development of medicine and life sciences, it provides an efficient technology solution to study disease and improve the ability of treatment.

III. The Principle of TCM for Hypertension

Hypertension is a complex chronic disease with high mortality and disability. BP regulation is related to complex interactions, such as the heart, kidneys, central nervous system, peripheral nervous system, and endothelial system, age, diet and living habits, emotional disorders, overtired, environment, and elderly body deficiency. The etiology and pathogenesis of hypertension are directly related to the syndrome differentiation and therapeutic methods [3]. In conventional medicine, there are two classifications of hypertensive disease: essential

Authorized licensed use limited to: Padre Conceicao College of Engineering. Downloaded on August 06,2024 at 09:21:45 UTC from IEEE Xplore. Restrictions apply.

hypertension and secondary hypertension. The clinical diagnosis is the measurement of BP and follow-up. Controlling BP levels by pharmacological treatment is the general method to delay the development of the disease and prevent damage to other target organs. Unfortunately, the etiology and pathogenesis of hypertension are not accurately distinguished, and the drug therapy cannot achieve the goal of a complete cure.

The TCM has been used to treat diseases in China for more than two thousand years and has historically established itself as a system of healthcare in China. It regards the human body as a whole, not as separated systems, focuses on prevention, whole regulation, and comprehensive intervention, and works better at low costs to manage common diseases. In TCM, hypertension disease is considered "vertigo", with typical clinical manifestations such as dizziness, headache, fatigue, and lassitude in the loins and knees [2]. According to the theory, the primary pathogenesis of vertigo is stagnation of phlegm, stagnation of blood, excess of the liver, and both liver and kidney deficiency [9]. Therefore, the treatment focuses more on pathogenesis identification and syndrome differentiation to relieve symptoms and recover viscera functions. From the perspective of TCM, doctors divide and treat hypertension into three syndrome types described below.

1) *Type I (Syndrome of Liver Yang Hyperactivity):* For exuberant people, under the influence of long-term emotional disaffection, depression, or anger, it is easy to become fire syndrome, which depletes the liver and eventually transforms into the imbalance of Yin and Yang. Therefore, emotional counseling and lifestyle modifications are essential. In addition to conventional treatment, Taichong, Xingjian, Dadun, and Ganshu acupoints therapy, Tianma Gouteng decoction, Zhengan Xifeng decoction, Longdan Xiegan decoction compound drugs could achieve a therapeutic effect.

2) *Type II (Stagnation of Phlegm-Dampness):* In an unsuitable diet case for the long term, it is easy to damage the patient's spleen ("pancreas" in anatomy) and weaken stomach function, which leads to spleen and stomach failure in dealing with water and nutrition, and then turn phlegm and dampness, furthermore impact lungs. Aerobic exercise should be recommended as a priority which could effectively improve blood circulation, enhance the function of the spleen and stomach, promote the metabolism of water and nutrients, and block sputum production. Thus, the comprehensive intervention of compound drugs, acupoints, and physical exercises will have significant curative effects.

3) *Type III (Syndrome of Kidney Yin Deficiency):* Kidney function decline is related to chronic illness, long-term fatigue, and age, which cause the kidney Yin and complicated with hyperactivity of Yang. Meditation and some kidney-yin nourishing foods, such as carp, oysters, and black chicken, help to improve renal function and balance the imbalance between kidney Yin and kidney Yang. Pill of Rehmannia, Qiju Dihuang pill, and Jinkui Shenqi Wan is the classical representative famous prescriptions for treating kidney deficiency and excessive fluid syndrome.

According to the TCM theory, symptom differentiation is the basis for diagnosis and treatment. Not only compound drug treatment but also acupuncture, lifestyle modifications, diet, physical activity, emotional counseling, meditation, and alternative medicines have beneficial effects on the treatment of hypertension. Therefore, analyzing common types of hypertension discussed above, treating patients according to the different syndrome, and providing a specific therapeutic schedule assist in improving the efficacy of conventional treatment.

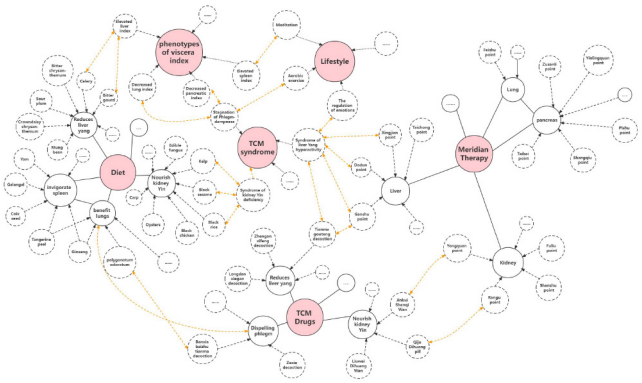

Fig. 3. Part of the knowledge relationship of the knowledge graph.

IV. A Case Study for Hypertension Patient Healthcare

In this paper, we construct a parallel healthcare system consisting of wearable TCM pulse pattern diagnostic devices, electronic sphygmomanometers, cellphone apps, and an automated intelligence platform, and performed under the supervision of real experienced doctors. In the TCM, skilled doctors use palpation to collect clinical information to diagnose syndrome patterns, distinguish pathogenesis, and draw conclusions. Nowadays, many devices for novel pulse sensing in clinical practice can record and classify human arterial pulses to provide more objective and quantitative data [10]. We use pulse diagnostic devices to detect and record the patient's viscera functional status through the decomposition and feature extraction of the pulse signals expressed by the calculated viscera index (heart, liver, kidneys, pancreas, and lungs). The BP and viscera indexes are considered the critical parameters for description patients that reveal the pathogenesis and etiology and help to distinguish disease syndrome types of patients in artificial systems. Under the principle of hypertension discussed in Section III, the computational experiments will be carried out with the combination of actual patients' status, clinical evidence, and doctors' experience. As shown in Fig. 3, we provide a medical knowledge network for computational experiments in hypertension diagnosis, treatment, and management. When a series set of parameters of a patient are recorded, different treatment schemes will be computed and simulated the effectiveness to recommend the best treatment scheme for the actual doctor.

There were 51 hypertension patients (26 male and 25 female) and 50 healthy people (26 male and 24 female) who volunteered to participate, aged from 27 to 82 (median age 55). Three months of experimental data were collected anonymously with the participators' authorization. Based on

Authorized licensed use limited to: Padre Conceicao College of Engineering. Downloaded on August 06,2024 at 09:21:45 UTC from IEEE Xplore. Restrictions apply.

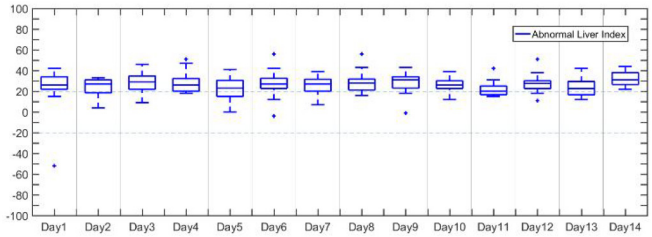

Fig. 4. Liver index of liver Yang hyperactivity.

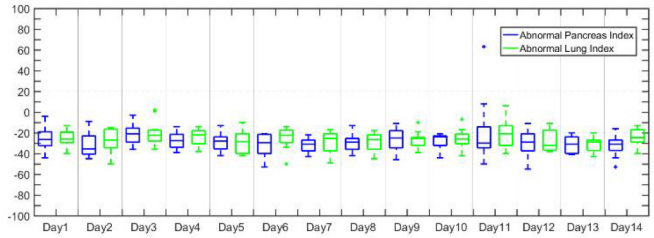

Fig. 5. Pancreas and lungs index of Stagnation of Phlegm-dampness.

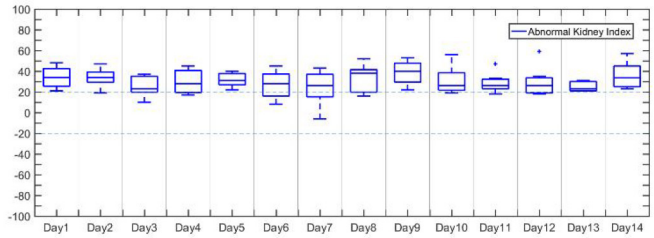

Fig. 6. Kidney index of Kidney Yin deficiency.

statistics and analysis, the viscera indexes positively correlate to the characteristics of different hypertension types. The viscera index ranges from -100 to $+100$, representing the functional status, which the negative value corresponding to deficiency and the positive value corresponding to hyperactivity. The data sample statistics of healthy people show that the baseline range is set from -20 to $+20$, which means functional normal, and balanced. Ordinary, healthy people's internal organs operate and interact harmoniously. Their viscera indexes vary slightly and fluctuate around a balance status baseline established from a large amount of accurate statistical data. On the contrary, the patient's syndrome index varies dramatically, and regular fluctuations away from the baseline correspond to the pathogenesis in TCM theory.

A. Type I Corresponds to the High Liver Index

As shown in Fig. 4. The liver indexes of patients are chronically high, generally exceeding a shift above the baseline of more than 20, which corresponds to the syndrome of liver Yang hyperactivity. The data indicates the pathogenesis caused by high metabolism, excessive liver Yang, insufficient convergence, and imbalance of Qi movement. Clinical manifestations include dizziness, headache, tinnitus, insomnia and dreams, irritability, fatigue, and emotional excitement tend to aggravate the condition.

B. Type II Corresponds to a Combination of Low Pancreas and Low Lung Index

Type II hypertension patients have the characteristics of pancreas and lungs indexes were both downward from baseline (shown in Fig. 5), with a median value of -20 and below, which corresponds to the phlegm-dampness syndrome. The causality for this hypertension type is mainly the poor function of the pancreas and stomach, which cannot effectively deal with phlegm and humidity, resulting in excessive phlegm in the body and further affecting lung function. Clinical manifestations include dizziness, headache, heavy head feeling, chest tightness, palpitation, indigestion, nausea, vomiting, slight edema, and weakness of lower limbs.

C. Type III Corresponds to a High Kidney Index

When the kidney index deviates upward from the baseline by more than 20 (shown in Fig. 6), it belongs to the type of high kidney index and corresponds to kidney-yin deficiency syndrome. This type of hypertension is mainly due to the decline of Yin qi in the human body, especially the deficiency of kidney Yin, which leads to the relative overabundance of Yang qi and eventually causes the imbalance of blood pressure regulation. Clinically, it shows multiple vertigo accompanied by night sweats, insomnia, forgetfulness, tinnitus, fatigue, waist and knees, and other symptoms.

With the classification of the syndrome type, target treatment options are proposed. In the actual patient and healthy volunteers experiment, the description represented by viscera indexes and BP is close to ordinary healthy people if the treatment is effective. Reversely, they remain unchanged if the treatment is ineffective.

V. Conclusion

This paper proposes a parallel framework for the hypertension healthcare system. First, we discussed the description, prediction, and prescriptiveness patients performed and interacted with each other. Then, the TCM method for three types of hypertension is reviewed, providing a particular viscera index that can be measured objectively. At last, we constructed a PHS for hypertension treatment following the TCM and proposed optional targeted treatment corresponding to each type. As a result, the PHS benefits hypertension patients in the actual treatment. In the future work, following issues will be continued: (a) structure the disease's treatment pathway and optimize treatment using the knowledge graph learning and inference, (b) conduct more evidence-based controlled clinical trials.

References

[1] T. P. Zonneveld et al., "Blood pressure-lowering treatment for preventing recurrent stroke, major vascular events, and dementia in patients with a history of stroke or transient ischaemic attack," *Cochrane Database Syst. Rev.*, no. 7, 2018, Art. no. CD007858.

[2] J. Wang and X. J. Xiong, "Control strategy on hypertension in Chinese medicine," *Evidence Based Complementary Alternative Med.*, vol. 2012, Dec. 2011, Art. no. 284847.

Authorized licensed use limited to: Padre Conceicao College of Engineering. Downloaded on August 06,2024 at 09:21:45 UTC from IEEE Xplore. Restrictions apply.

[3] T. Unger et al., "2020 international society of hypertension global hypertension practice guidelines," *Hypertension*, vol. 75, no. 6, pp. 1334–1357, 2020.
[4] F. Y. Wang, "Parallel system methods for management and control of complex systems," *Control Decis.*, vol. 19, no. 5, pp. 485–489, 2004.
[5] F. Y. Wang, "Parallel control: A method for data-driven and computational control," *Acta Automatica Sinica*, vol. 39, no. 4, pp. 293–302, 2013.
[6] F.-Y. Wang and P. K. Wong, "Intelligent systems and technology for integrative and predictive medicine: An ACP approach," *ACM Trans. Intell. Syst. Technol.*, vol. 4, no. 2, pp. 1–6, 2013.
[7] S. Wang et al., "Blockchain-powered parallel healthcare systems based on the ACP approach," *IEEE Trans. Computat. Soc. Syst.*, vol. 5, no. 4, pp. 942–950, Dec. 2018.
[8] X. Wang, L. X. Li, Y. Yuan, P. J. Ye, and F.-Y. Wang, "ACP-based social computing and parallel intelligence: Societies 5.0 and beyond," *CAAI Trans. Intell. Technol.*, vol. 1, no. 4, pp. 377–393, 2016.
[9] L. S. Liu et al., "Chinese guidelines for prevention and treatment of hypertension 2018 revised edition," *Prevent. Treatment Cardio Cereb. Vasc. Dis.*, vol. 19, no. 1, pp. 1–44, 2018.
[10] K. W. Kong et al., "Sphygmopalpation using tactile robotic fingers reveals fundamental arterial pulse patterns," *IEEE Access*, vol. 10, pp. 12252–12261, 2022.

Lin Ge received the bachelor's degree in electronic engineering and the master's degree in electromagnetic field and microwave technology from Northwestern Polytechnical University, Xi'an, China, in 2004 and 2007, respectively. He is currently pursuing the Ph.D. degree with the Faculty of Innovation Engineering, Macau University of Science and Technology, Macau, China. He is a Senior Engineer and the Director with the Research and Development Department, Sungohealth Inc., Beijing, China. His research interests include artificial intelligence, complex system, parallel medicine, machine learning, and smart wearable devices.

Bo Lv received the bachelor's degree in traditional chinese medicine from the Beijing University of Chinese Medicine, Beijing, China, in 2015. He is an Assistant Researcher with the Institute of Life Science, Sungohealth Inc., Beijing. His current research interests include quantitative representation of TCM concepts and digital health assessment.

Na Li received the bachelor's degree in computer science and technology from Shanghai Jiao Tong University, Shanghai, China, in 2003. She is currently pursuing the M.B.A. degree with the Chinese Academy of Social Sciences. She is an Assistant Researcher with the Institute of Life Science, Sungohealth Inc., Beijing, China. Her current research interests include digital health, big data, and artificial intelligence.

Shan An (Senior Member, IEEE) received the bachelor's degree in automation engineering from Tianjin University, China, in 2007, and the master's degree in control science from Shandong University, China, in 2010. He is currently pursuing the Ph.D. degree with School of Computer Science and Engineering, Beihang University. His research interests include computer vision in health, AR and robotics, image retrieval, and image segmentation. He has served as a Program Committee Member for ACM Multimedia (2019–2022), AAAI (2022), and IJCAI (2021–2024). He is a Reviewer of more than twenty highly prestigious journals and conference papers, such as IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON MULTIMEDIA, and *Pattern Recognition*, CVPR, ICCV, and ICRA.

Fei-Yue Wang (Fellow, IEEE) received the Ph.D. degree in computer and systems engineering from Rensselaer Polytechnic Institute, Troy, NY, USA, in 1990.

He joined The University of Arizona in 1990, and became a Professor and the Director of the Robotics and Automation Laboratory and the Program in Advanced Research for Complex Systems. In 1999, he founded the Intelligent Control and Systems Engineering Center with the Institute of Automation, Chinese Academy of Sciences (CAS), Beijing, China, under the support of the Outstanding Chinese Talents Program from the State Planning Council and in 2002, was appointed as the Director of the Key Laboratory of Complex Systems and Intelligence Science, CAS, and the Vice President of Institute of Automation, CAS in 2006. He found CAS Center for Social Computing and Parallel Management in 2008, and became the State Specially Appointed Expert and the Founding Director of the State Key Laboratory for Management and Control of Complex Systems in 2011. His current research focuses on methods and applications for parallel intelligence, social computing, and knowledge automation.

Dr. Wang received the National Prize in Natural Sciences of China, Numerous Best Papers Awards from IEEE Transactions and became an Outstanding Scientist of ACM for his work in intelligent control and social computing in 2007, and received the IEEE ITS Outstanding Application and Research Awards in 2009, 2011, and 2015, respectively, the IEEE SMC Norbert Wiener Award in 2014, and became the IFAC Pavel J. Nowacki Distinguished Lecturer in 2021. Since 1997, he has been serving as the General or the Program Chair of over 30 IEEE, INFORMS, IFAC, ACM, and ASME conferences. He was the President of the IEEE ITS Society from 2005 to 2007, the Chinese Association for Science and Technology, USA, in 2005, the American Zhu Kezhen Education Foundation from 2007 to 2008, the Vice President of the ACM China Council from 2010 to 2011, the Vice President and the Secretary General of the Chinese Association of Automation from 2008 to 2018, the Vice President of IEEE Systems, Man, and Cybernetics Society from 2019 to 2021, and the IEEE Council of RFID from 2019 to 2021. He was the Founding Editor-in-Chief (EiC) of the *International Journal of Intelligent Control and Systems* from 1995 to 2000, *IEEE Intelligent Transportation Systems Magazine* from 2006 to 2007, IEEE/CAA JOURNAL OF AUTOMATICA SINICA from 2014 to 2017, Journal of Command and Control (China) from 2015 to 2021, and *Journal of Intelligent Science and Technology* (China) from 2019 to 2021. He was the EiC of the IEEE INTELLIGENT SYSTEMS from 2009 to 2012, IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS from 2009 to 2016, IEEE TRANSACTIONS ON COMPUTATIONAL SOCIAL SYSTEMS from 2017 to 2020. He is currently the President of CAA's Supervision Council and the EiC of IEEE TRANSACTION ON INTELLIGENT VEHICLES. He is a Fellow of INCOSE, IFAC, ASME, and AAAS.

Authorized licensed use limited to: Padre Conceicao College of Engineering. Downloaded on August 06,2024 at 09:21:45 UTC from IEEE Xplore. Restrictions apply.